ECEN 133 HW 5 Dylan Thomas

1.

a. $X[k] = \sum_{n=0}^{N-1} x(n) \cdot e^{-j2\pi kn/N}$ $X_a = e^{-i2\pi(0)3/4} + e^{-i2\pi(0)2/4} + e^{-2\pi(0)(1/4)} + e^{-2\pi(0)}$

$= 4$

repeat for $k=1,2,3$ and all $= 0$

$X_a = (4,0,0,0)$ phase $= (0,0,0,0)$ (no imaginary part)

mag $= (4,0,0,0)$

$X_b = e^{-i2\pi(0)0/4} - e^{-i2\pi(0)1/4} + e^{-i2\pi(0)2/4} - e^{-i2\pi 3/4} = 0$

repeat for $k=1,2,3$ and get $(0,-4,0)$

$X_b = (0,0,4,0)$ phase $= (0,0,0,0)$ (no imaginary part)

mag $= (0,0,4,0)$

$X_c = e^{-i2\pi(0)(0/4)} + e^{-i2\pi(0)2/4} = 2$

repeat for $k=1,2,3$ and get $(0,2,0)$

$X_c = (2,0,2,0)$ phase $= (0,0,0,0)$ (no imaginary part)

mag $= (2,0,2,0)$

b.

$F_s = 1000$

$(0, 250, 500, 750)$

X_a has freq 0 Hz

X_b has freq 500 Hz

X_c has freqs 0 Hz and 500 Hz

c. $\frac{X_a + X_b}{2} = X_c$

$\frac{(1,1,1,1) + (1,-1,1,-1)}{2} = (1,0,1,0)$

and

$\frac{(4,0,0,0) + (0,0,4,0)}{2} = (2,0,2,0)$

so

$\frac{X_a + X_b}{2} = X_c$ ✓

matrix: $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$

2. $x_a = [1, 1, -1, -1]$

a.

$1+1-1-1=0$

$1-j+1-j=2-2j$

$1-1-1+1=0$

$1+j+1+j=2+2j$

$$\boxed{\begin{aligned} X_A &= [0, 2-2j, 0, 2+2j] \\ \text{mag} &= [0, 2\sqrt{2}, 0, 2\sqrt{2}] \\ \text{phase} &= [0, -\pi/4, 0, \pi/4] \end{aligned}}$$

b. $x_b = [1, -1, -1, 1]$

$1-1-1+1=0$

$1+j+1+j=2+2j$

$1+1-1-1=0$

$1-j+1-j=2-2j$

$$\boxed{\begin{aligned} X_b &= [0, 2+2j, 0, 2-2j] \\ \text{mag} &= [0, 2\sqrt{2}, 0, 2\sqrt{2}] \\ \text{phase} &= [0, \tfrac{\pi}{4}, 0, -\tfrac{\pi}{4}] \end{aligned}}$$

x_c I can see $\frac{x_a + x_b}{2} = x_c$

again, so $\frac{x_a + x_b}{2} = x_c$

$$X_c = \frac{[0, 2-2j, 0, 2+2j] + [0, 2+2j, 0, 2-2j]}{2} =$$

$$\boxed{\begin{aligned} &[0, 2, 0, 2] \\ \text{mag} &= [0, 2, 0, 2] \\ \text{phase} &= [0, 0, 0, 0] \end{aligned}}$$

b. All signals have 250 and 750 Hz present.

c.

X_b and X_a have the same freqs but at diff phases. This is b/c $x_a = x_b$ shifted right by 1

We can see this w/ the circular time shift property.

$$x(n-m) \bmod N = X(k) e^{-j2\pi k m / N}$$

$$X_A(1) = X_b(1) e^{-j2\pi(1)(1)/4}$$

$$2-2j = (2+2j) e^{-j\pi/2} = 2-2j \checkmark$$

d.

$$X_c = \frac{X_a + X_b}{2}$$

4.

a. Freq res $= \frac{F_T}{N}$

$\frac{2000}{50} = 40Hz$ $\frac{4000}{50} = 80Hz$ $\frac{1000}{50} = 20Hz$

$\frac{2000}{100} = 20Hz$ $\frac{4000}{100} = 40Hz$ $\frac{1000}{100} = 10Hz$

b. cycles $= \frac{N}{F_T} f_0$ $(f_0 = 220)$

$220\left(\frac{50}{2000}\right) = 5.5$ cycles $\frac{50}{4000}(220) = 2.75$ cycles $220\left(\frac{50}{1000}\right) = 11$ cycles

$220\left(\frac{100}{2000}\right) = 11$ cycles $\frac{100}{4000}(220) = 5.5$ cycles $220\left(\frac{100}{1000}\right) = 22$ cycles

If the cycles are an int, the peak amplitude is one clean spike with no spectral leakage.

c. time $= \frac{1}{\text{freq res}}$

$\frac{1}{40} = .025s$ $\frac{1}{80} = .0125s$ $\frac{1}{20} = .05s$

$\frac{1}{20} = .05s$ $\frac{1}{40} = .025s$ $\frac{1}{10} = .1s$

longer sequences give better (smaller) freq res which means bins are smaller and samples are closer together.